

LAB 1.2: Incremental Encoder

1 fig 00

- `in07A0f8 d57340500014`
- `in09f0f11 gnd0f57340500031`
- `in09f0f11H00 000f0067340500073`

š3 fl7i cõ (Objectives)

1. `š2 2090791300BoF7ñ4i šý 9F304` Quadrature Signals) `š1š13šğ ùcõA00708007`
`f0šnoÁfñ šffñš0šfj 4Phase Relationship)š2š14šfš4CWš07hÁšfñ4CCW)š`
2. `š2 209ý290Over} owš 9fÁ05)š4cõ hardware Counterš16-bitš0744i 9tš4š 4f0š0`
`Wrap-aroundš2š9i šý290š0š 9fñš0š2ñš 71fñ Accumulated Count)š0š4š3š0š`
3. `š2š304cõš070%ÁF290š0 Pulses Per Revolution (PPR)š4cõ Encoderš0š0š070%ÁF290š`
`ôÁfñ0š4fš0šfñ4 Angular Resolution)š4š7tšš`
4. `š2š30cõš Raw Countsš1ššñi 9fñš0Over} owš0š0š2šñšÁfñš0Á fffñš0š 0š2šñš`
`š0šfñ4 Radians)š070Áfñš0šfñ4 rad/s)š`
5. `š2š44i 9tš907ñšfñ š0šbming Sequenceš 9fš0ššñš0šš0š`
6. `š2š0šfñ729073 fljt šfljt 100šfñš40šfš 9fñš3fñš0š (Decoding Modes)štšš0š1š0š2š0š4š0š`
7. `š2š0šfñ72š0š fljtšñ4cõš9š0šñi 9fñš4fñš0š4fñšfñš4šý 9fñš0šš0šfñ4š`

Incremental Encoder

Incremental Encoder f_{inc}: g_t → h_t, n_t → h_t (Squared Wave) 2048
g_t → h_t, n_t → h_t

- $\dot{\theta} = \frac{1}{r} \frac{dr}{dt} \dot{\phi}$ (Relative): Encoder 3, 7, 8, 11, 13, 14, 16, 17, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840

[illegible]

- X1 Single-edge counting: $\frac{1}{n} \sum_{i=1}^n \frac{d_i}{d_i + 1}$
- X2 Double-edge counting: $\frac{1}{n} \sum_{i=1}^n \frac{d_i^2}{d_i^2 + 1}$
- X4 Quadrature decoding: $\frac{1}{n} \sum_{i=1}^n \frac{d_i^4}{d_i^4 + 1}$

3. 3y290Over} ovr0740i 4fB2dWrap-around

0ni 9fh3040nH-bredTimer0103n9t t 06-bit02 f9f0ÁfiÁg0ÁB0(Counter Register)0
19f9fh00930538fj02¹⁶ - 1 = 65535

- Over} ovr0740i 4fB2dWrap-around0655350nfh030097Áni 0B0Polls0ver)09fB0nfh00
0fh0
- Under} ovr0740i 4fB2dWrap-around0655350nfh030097Áni 0B0h065535
- Wrap-around0655350nfh030097Áni 4fB2dWrap-around0655350nfh030097Áni 9t t f9f0ÁfiÁg0ÁB0(Counter Register)0
(8n09 655340302f0203065534)007h09f038fj0sh00f 009 d0Á0nd Á 2f0
0t 3Áf065536)0f02030092h071f0(Accumulated Count)0fh0300

ì 9f00959fB9Áf0(OC0setup)

8f0209 9f9fh03040078fj0t 8fj0t 0fh0i 9fh43fh0sh0d 000 8030fh0i 9f00959fB9Áf0(OC0setup)
04f09Áf05TMB2Cube0C00000

1. TIMB010%fb02fh030X4):

- 8d010000i 00TIMB010%fb02fh030X4)0General-Purpose0Timer0fh04fB0Encoder0Mode
- ì 9f00959fB9Áf0(OC0setup)02fh030Encoder0Mode0T10and0T200ni 9f00959fB9Áf0(OC0setup)09f0T100
(PA4)0070T20(PA6)0fh0sh0h04t 0409f0T0A0070B0X40Decoding0
- Counter0Period0(ARR):0009Áf0655350030n09053040016-bit08f020fh0Á0 9fh0sh0Á0
fh030i 0n0800Over} ovr0740i 4fB2dWrap-around

2. TIM1010%fb02fh030X2):

- 8d010000i 00TIM1010%fb02fh030X2)0Advanced-Control0Timer0
- ì 9f00959fB9Áf0(OC0setup)02fh030Encoder0Mode0T1009fB9Áf0(OC0setup)0n0sh0h04t 0n0707n09000409f0T0h0
040T10(PA8)0fh0sh0h04t 0T20(PC1)07008f0i 0%h03fh0h00000Áf07080fj09t t 0X20

- 18 y 9 f Paw Counts 791 3 off 1 diff 0 2 off 1 2 off 1 (Sawtooth) 25 g 18 0 8 1 65535
- 48 4 f 122 Wap-around 9 f 9 h 1 f 5 8 i 9 B 30 Over } own 40 f Counts 3 g i 9 f 10 f 1 h 9 0 f 0 9 i 1 f 5 1 f 5 0 9 2 i 1 71 f 1 5 4 8 3 0 3 9

3. $d_{3f}^{\frac{1}{2}}$

- $\text{db}_3 \text{ fl} \hat{r} \hat{o} \hat{g} \hat{f} \hat{i} \hat{A} \hat{e} \hat{n} \hat{c} \hat{o} \hat{d} \hat{e} \hat{r} \hat{o} \hat{f} \hat{s} \hat{o} \hat{d} \hat{a} \hat{s} \hat{o}$
- $\text{db}_3 \text{ fl} \hat{r} \hat{o} \hat{g} \hat{f} \hat{i} \hat{P} \hat{a} \hat{w} \hat{d} \hat{C} \hat{o} \hat{u} \hat{n} \hat{t} \hat{s} \hat{g} \hat{r} \hat{i} \hat{M} \hat{B} \hat{),} \hat{A} \hat{c} \hat{c} \hat{u} \hat{m} \hat{u} \hat{l} \hat{a} \hat{t} \hat{e} \hat{d} \hat{C} \hat{o} \hat{u} \hat{n} \hat{t} \hat{s} \hat{g} \hat{r} \hat{W} \hat{r} \hat{a} \hat{p} \hat{a} \hat{r} \hat{o} \hat{u} \hat{n} \hat{d} \hat{),} \hat{A} \hat{n} \hat{g} \hat{u} \hat{l} \hat{a} \hat{r} \hat{e} \hat{P} \hat{o} \hat{s} \hat{i} \hat{t} \hat{i} \hat{o} \hat{n} \hat{s} \hat{g} \hat{r} \hat{a} \hat{d} \hat{),} \hat{A} \hat{n} \hat{g} \hat{u} \hat{l} \hat{a} \hat{r} \hat{e} \hat{V} \hat{e} \hat{l} \hat{o} \hat{c} \hat{i} \hat{t} \hat{y} \hat{g} \hat{r} \hat{a} \hat{d} \hat{/} \hat{s}$
- $\text{db}_3 \text{ fl} \hat{A} \hat{t} \hat{t} \hat{e} \hat{r} \hat{C} \hat{o} \hat{u} \hat{n} \hat{t} \hat{e} \hat{r} \hat{P} \hat{e} \hat{r} \hat{i} \hat{o} \hat{d} \hat{g} \hat{r} \hat{6} \hat{5} \hat{5} \hat{3} \hat{5}$

4. $\tilde{n} \tilde{a} d_4 n_i$ $g \beta \% \delta \tilde{a} \tilde{o} g_n$

1. `float pos = 0;` (Initial Position in radians)
2. `int totalCounts = 0;` (Total Counts for one full revolution)
3. `int accumulatedCount = 0;` (Accumulated Counts)
4. $Position(rad) = AccumulatedCount * \left(\frac{2\pi}{Total\ Counts\ per\ Revolution} \right)$
5. $Velocity\left(\frac{rad}{s}\right) = \frac{d(Position)}{dt}$ (Derivative)
6. `pos = pos + (float) accumulatedCount / totalCounts * 2 * M_PI;` (Update Position)

5. 1.0i 9f h304ã

- [illegible]

3. $\tilde{n} \tilde{a} d' n i \text{ } g f B \% \tilde{o} n \tilde{o} g n$

- #### 4. Bài 304

5. 1f4l0i gfh304õ

- 9

2f9fjôÁfÁgôñh43õ9f2fñ43.754õ9õ78õ9f9f30fñ930õ4õõ9A Accumulated Countõ
8fñõ1õfõ

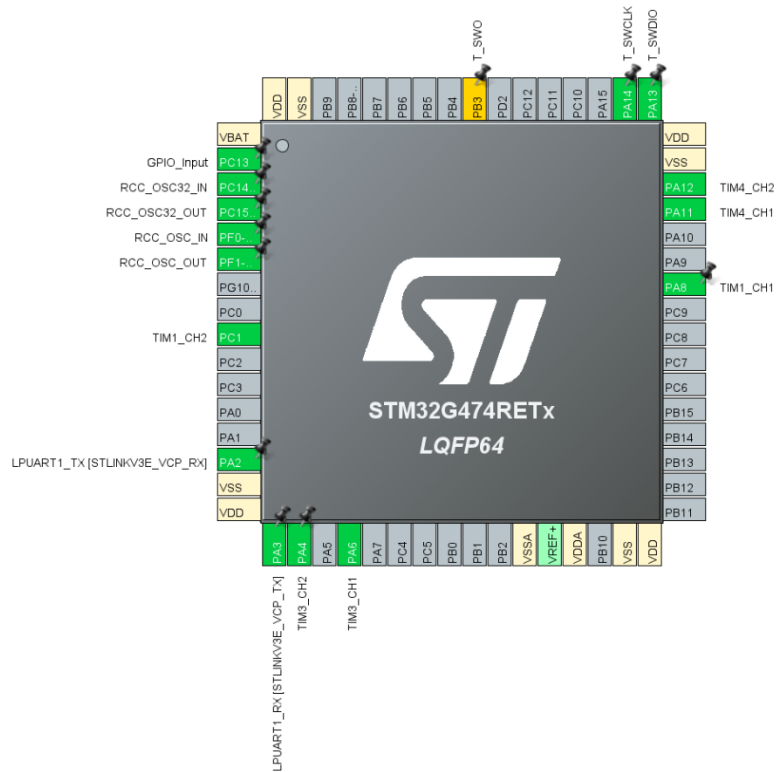
2. $\vec{r}(t)$ and $\vec{v}(t)$ (Position and Velocity)

ôÁf078fj3h96(Counts)ôñî1J03fjfd40BoT7ñ4ô f9ffñBâñ53fjtì8Encoderô
ôÁf078fj315(192Counts)ôñîñ gñ304c20B

- $\dot{\theta}$ (Angular Position):
 - BOUNS (240 PPR): $\dot{\theta}$ (Coarse Steps) $\approx 1.5^\circ$ (i.e. $1.5^\circ \times 240 = 360^\circ$)
 - CU (2048 PPR): $\dot{\theta}$ (Smooth Ramp) $\approx 0.17^\circ$ (i.e. $0.17^\circ \times 2048 = 360^\circ$)
- $\dot{\theta}$ (Angular Velocity):
 - BOUNS (240 PPR): $\dot{\theta}$ (Count 2048) $\approx 1.5^\circ$ (i.e. $1.5^\circ \times 240 = 360^\circ$)
 - CU (2048 PPR): $\dot{\theta}$ (Scattered Spikes) $\approx 0.17^\circ$ (i.e. $0.17^\circ \times 2048 = 360^\circ$)

ផ្ទាល់

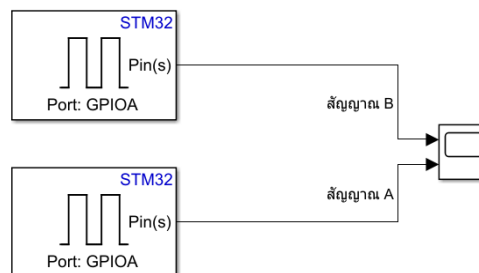
1. ទិន្នន័យ Pinout ឡើង



2. ទិន្នន័យ %405 Simulink

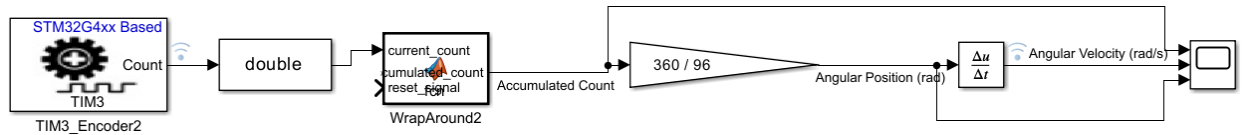
- ផ្ទាល់ Simulink Model ឡើង %405 ទិន្នន័យ %405 (Experiment) ឡើង

Reading Raw Signals



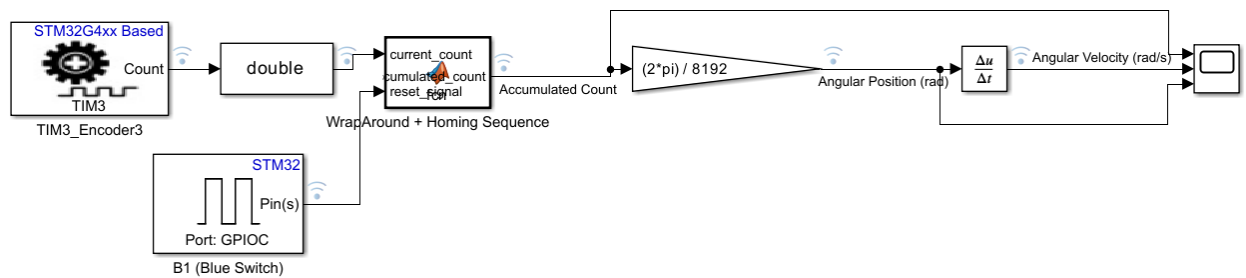
- Figure 14: Simulink Model of a Motor with a 360° Wrap-around (Experiment 2)

Angular Position (Radians)



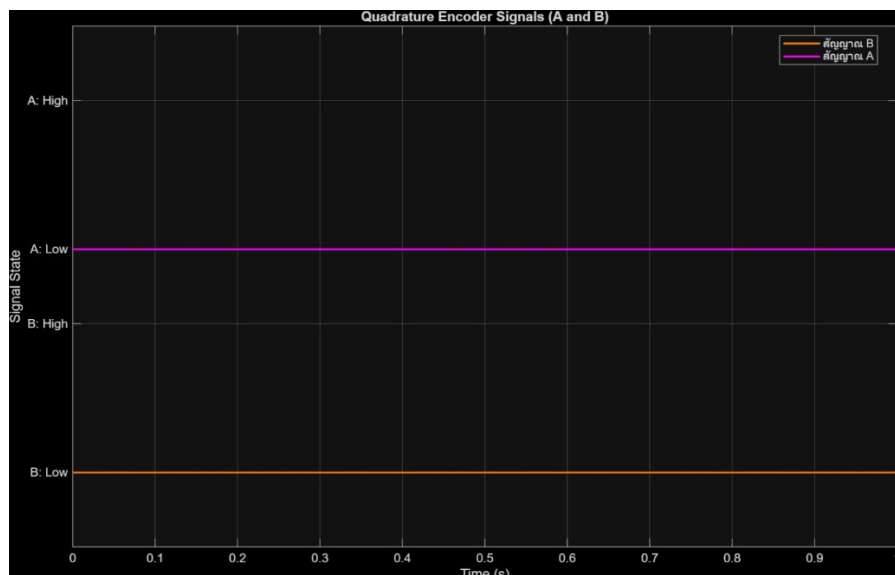
- Figure 15: Simulink Model of a Motor with a Homing Sequence (Experiment 4)

Homing Sequence

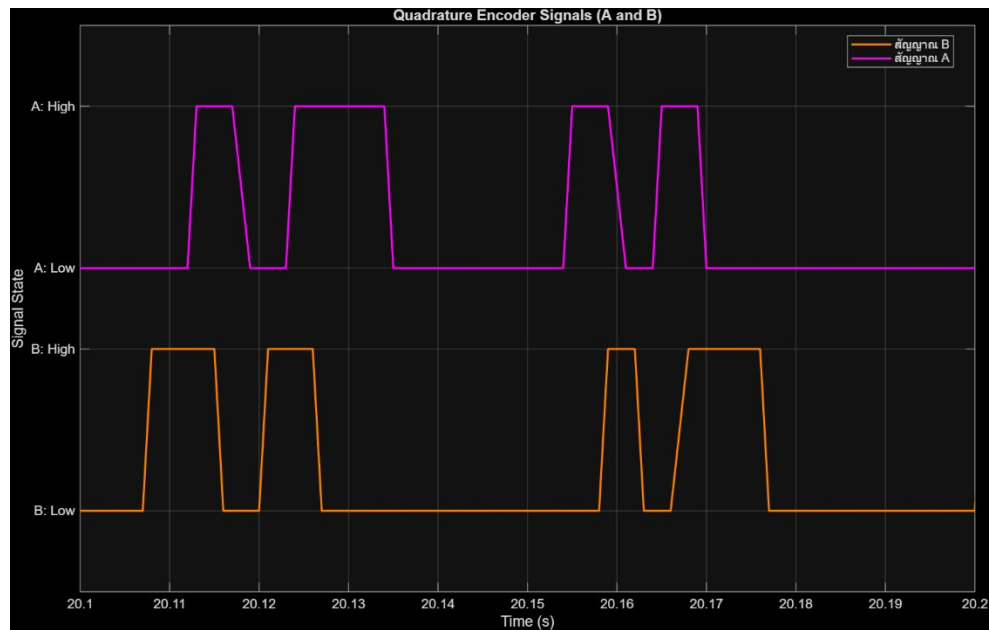


3.0 ffflD gh30d

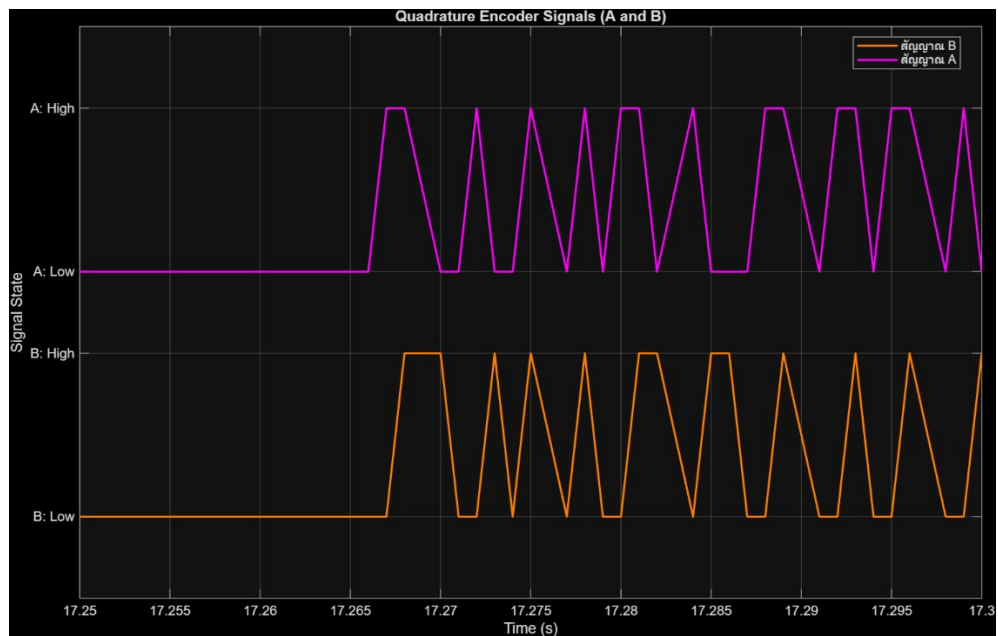
- Figure 16: Scope of the Motor with a Homing Sequence



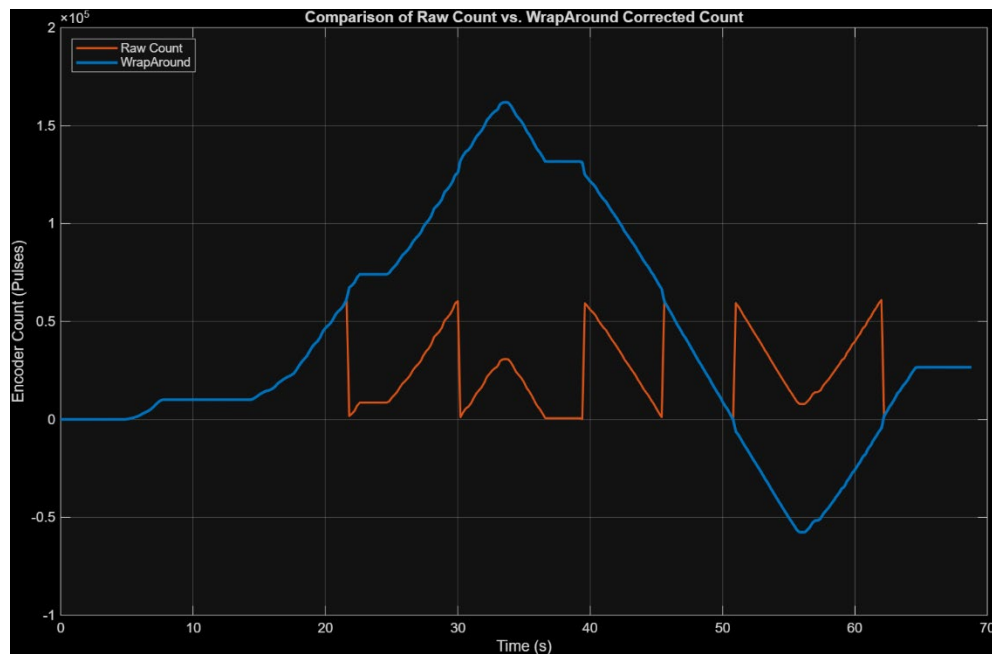
- $\frac{1}{2} \pi$ (90°) for A → B and $\frac{1}{2} \pi$ (90°) for B → A



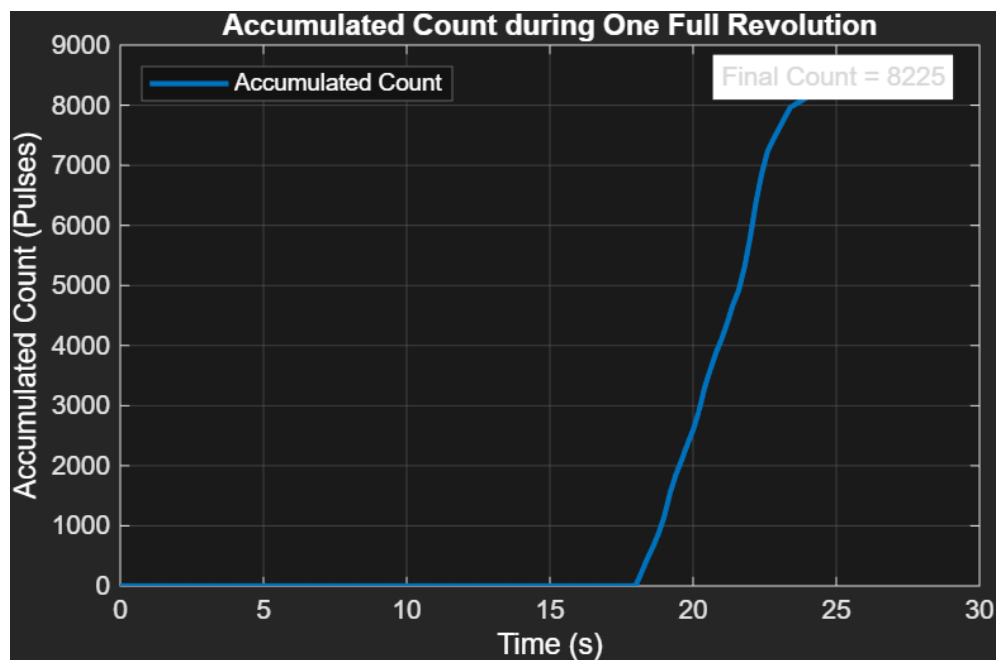
- $\frac{1}{2} \pi$ (90°) for A → B and $\frac{1}{2} \pi$ (90°) for B → A



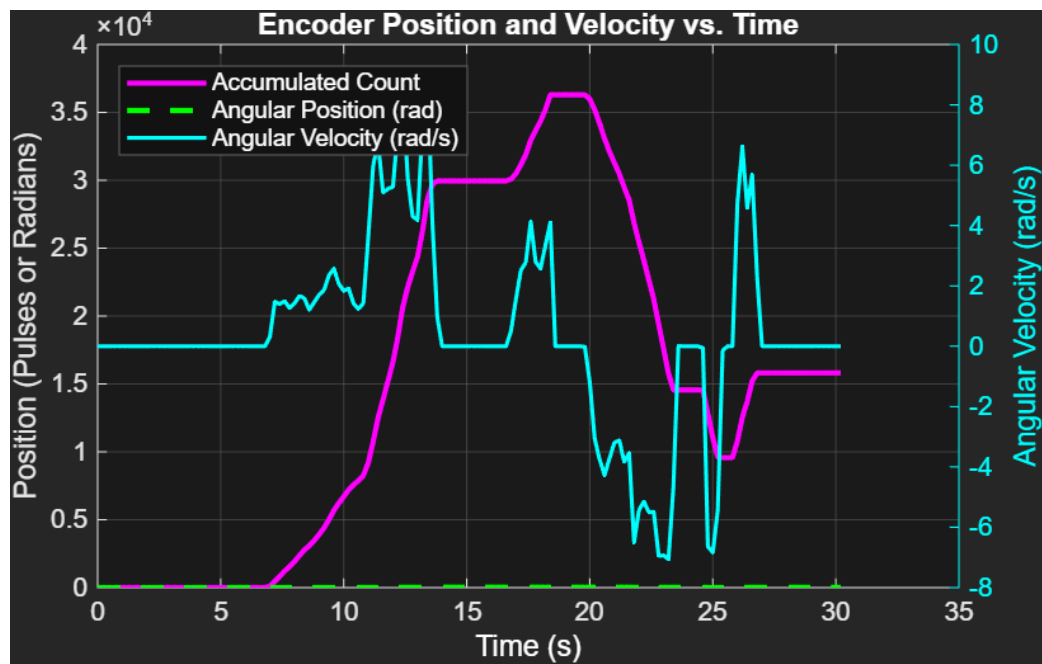
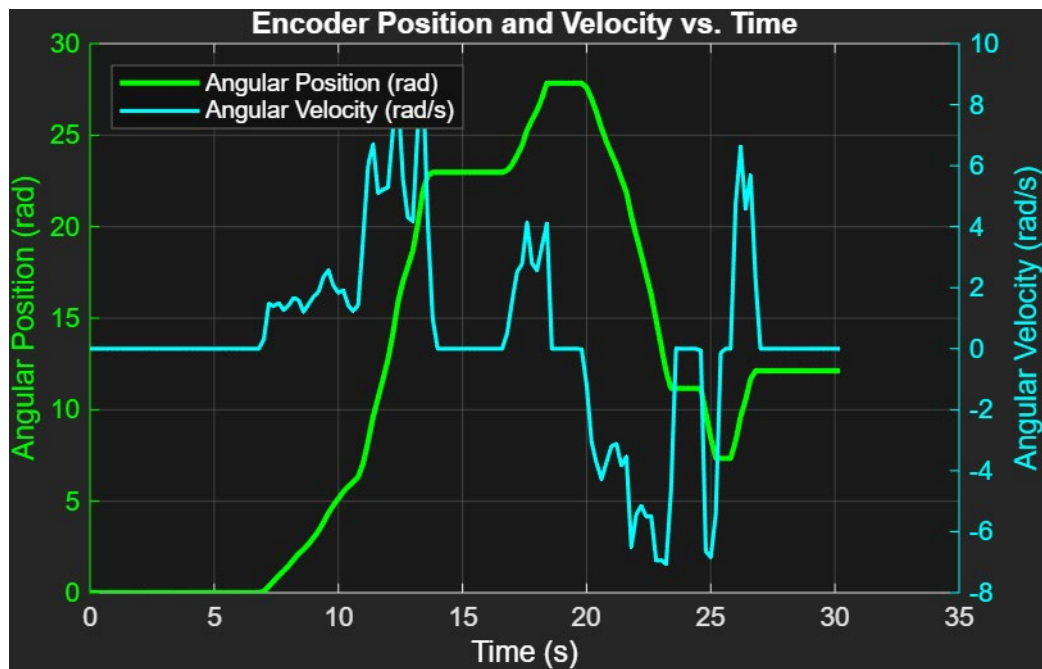
- $\int_{0}^{15} f(t) dt = 15$ Raw Count (1825) Accumulated Count (1825)



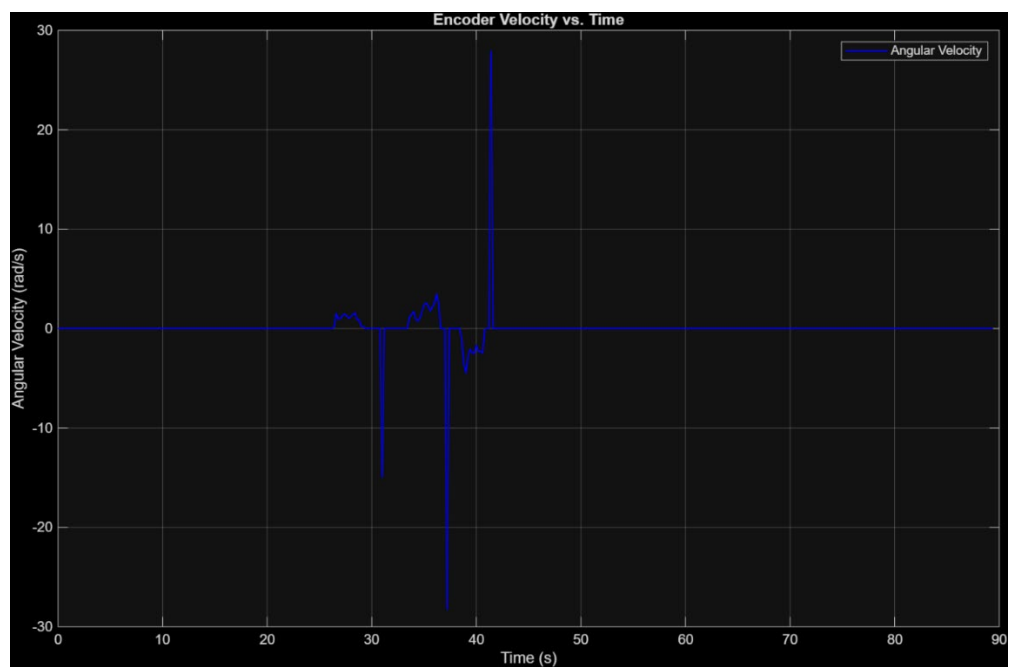
- $\int_{15}^{30} f(t) dt = 15$ Accumulated Count (1825) Final Count (8225)



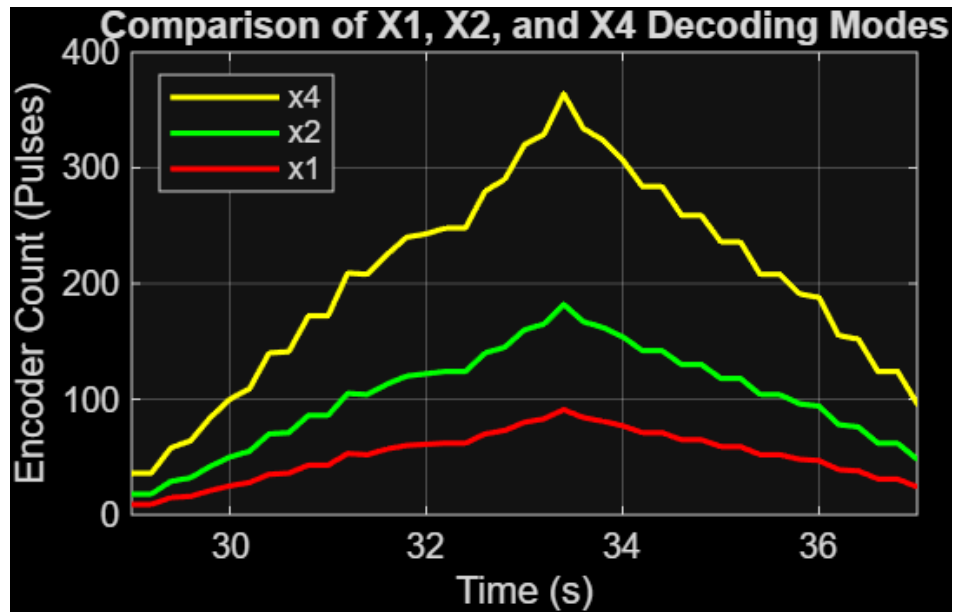
- $\dot{\theta} = \frac{d\theta}{dt}$ Position (rad) $\rightarrow$ Velocity (rad/s)



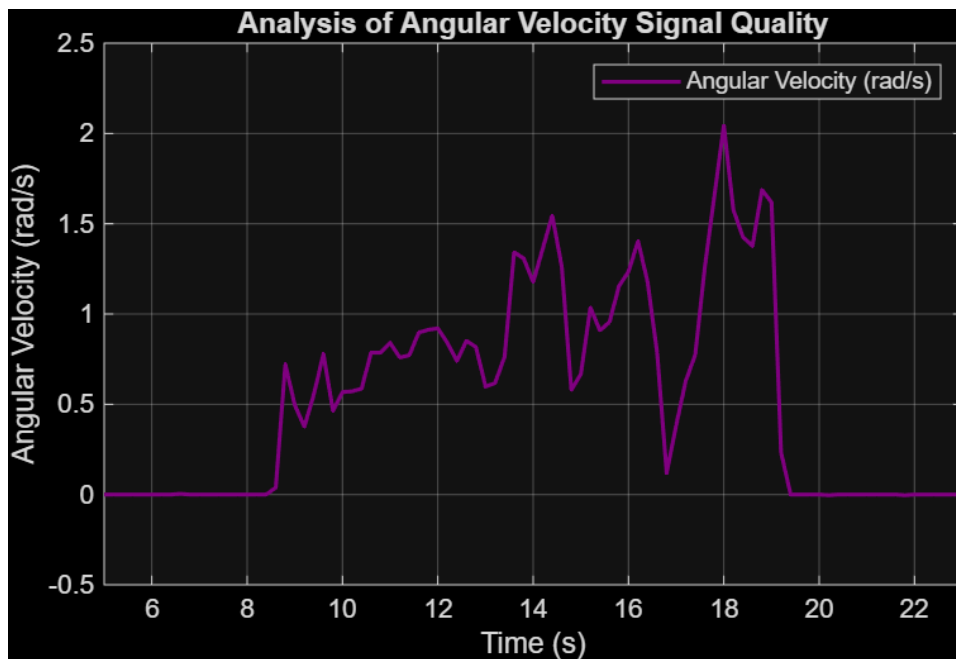
- $\dot{\theta} = 18.0 \text{ rad/s}$ Position 0.7 rad Velocity 2.0 rad/s Home



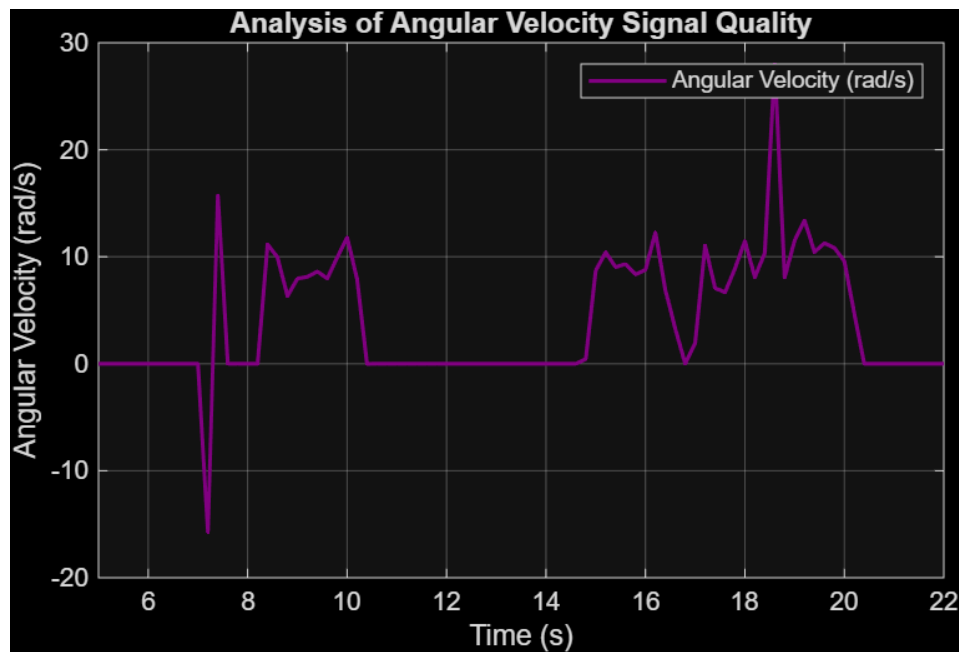
- `ifftshift` - Shifts the zero frequency component to the center of the spectrum.



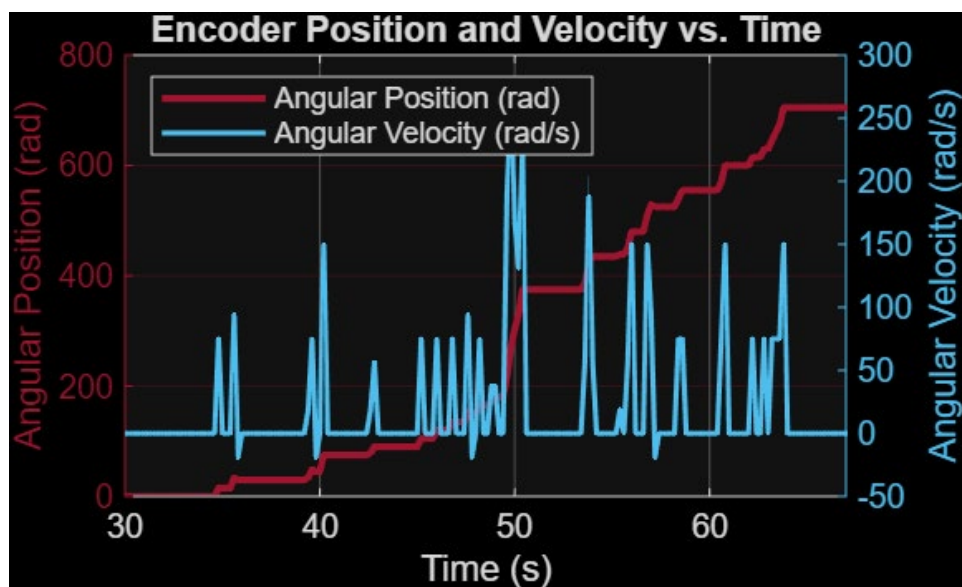
- `fftshift` - Shifts the zero frequency component to the center of the spectrum.



- $\dot{\theta} = 12.6 \text{ rad/s}$ (Velocity of 72 fms/10.35 spikes/Noise)



- $\dot{\theta} = 13.0 \text{ rad/s}$ (Encoder BURN 5124 PPR)



4.3.1 MATLAB Function (Wrap-around)

```
function accumulated_count_of_n(current_count)
persistent last_count;
persistent total_count;

% if n is 0, then % n is 0
if isempty(last_count)
    last_count = current_count;
    total_count = current_count;
end

% if n is 1
COUNTER_MAX = 65535;
COUNTER_HALF = 32768;

% if n is 2
delta = current_count - last_count;

% if n is 14
if delta > COUNTER_HALF
    if delta < 0
        if delta < -65534
            delta = -65533
        else
            delta = (COUNTER_MAX + 1);
        end
    else
        if delta > 65534
            delta = 65533
        else
            delta = (COUNTER_MAX + 1);
        end
    end
end

% if n is 17
total_count = total_count + delta;

% if n is 3
last_count = current_count;
```

%~~0~~ 08

accumulated_count~~total_count~~;

end

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- SCMA Co., Ltd.: Encoder (8410384f0247f1208) 9f49540f?!

<https://scma.co.th/blog/post/Encoder>

- PLC Automation: 08f1f04 incremental Encoder

<https://ploautomation.com/%E0%B9%82%E0%B8%84%E0%B8%A3%E0%B8%87%E0%B8%AA%E0%B8%A3%E0%B9%89%E0%B8%B2%E0%B8%87-incremental-encoder/>

- WJ Tech: Encoder (8410384f0247f1208) 9f49540f

<http://wjtech.co.th/activity/?p=2797>

- CUI Devices: AMT103-V Datasheet

https://www.alldatasheet.com/view.jsp?Searchword=Am103-v%20datasheet&gad_source=1&gad_campaignid=163458844&gclid=OAAAAADcdDU_DBveckKaMshlpPYxiEYY&gclid=GjwKCAjw04HBhB8EiwA8jGNbZ0C2Lk2ZZo45P8PaqjEFbPalrCINhSEEtMpznuQLDXnSj_HDWe-BoC2AKQAMD_BwE

- BOURNS: PEC11R-4220F-N0024

https://www.alldatasheet.com/view.jsp?Searchword=Pec11r&gad_source=1&gad_campaignid=170327939&gclid=OAAAAADcdDU-D4n4uDum6X06w_np8-EtV&gclid=GjwKCAjw04HBhB8EiwA8jGNbZQ_rUuVBpeSuedBqZeMA5xknW6ZUEy1jYJ76H1owhEFb3ILGaMkoCGSIQAMD_BwE